

p-BLOCK ELEMENTS

BORON FAMILY

- Order of Relative stability of +1 oxidation state in Boron family
 (1) $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$
 (2) $\text{Al} > \text{Ga} > \text{In} < \text{Tl}$
 (3) $\text{Al} = \text{Ga} = \text{In} = \text{Tl}$
 (4) $\text{Al} < \text{Ga} > \text{In} < \text{Tl}$
- Which of the following species act as oxidising agent?
 (1) Al^{+3} (2) Tl^{+3}
 (3) In^{+3} (4) Ga^{+3}
- Trihalide of boron family act as lewis acid. What is the order of lewis acidic strength?
 (1) $\text{BCl}_3 < \text{AlCl}_3 < \text{GaCl}_3$
 (2) $\text{BCl}_3 > \text{AlCl}_3 > \text{GaCl}_3$
 (3) $\text{BCl}_3 < \text{AlCl}_3 > \text{GaCl}_3$
 (4) $\text{BCl}_3 = \text{AlCl}_3 = \text{GaCl}_3$
- Which of the following oxide gives acidic solution in water?
 (1) B_2O_3 (2) Al_2O_3
 (3) Both (4) None
- White fumes appear around the bottle of anhydrous AlCl_3 . It is due to
 (1) Formation of $\text{Al}(\text{OH})_3$
 (2) Dimerisation of AlCl_3
 (3) Due to hydrolysis of AlCl_3 giving HCl gas
 (4) Due to absorption of moisture
- Borax dissolves in water to give
 (1) acidic solution (2) Basic solution
 (3) neutral solution (4) None
- In Borax bead test characteristic colour appears due to formation of—
 (1) Metaborate of transition metals
 (2) Sodium Metaborate
 (3) Boron oxide
 (4) Boric anhydride
- Which of the following statement is wrong about orthoboric acid
 (1) It has a layer structure
 (2) In its structure BO_3 units are bonded through H-bond
 (3) Boric acid is tribasic acid
 (4) On open heating for long time, it produces B_2O_3 .

- Diborane reacts with ammonia initially it gives $\text{B}_2\text{H}_6 \cdot 2\text{NH}_3$ which is formulated as—
 (1) $\text{BH}_3 \leftarrow \text{NH}_3$
 (2) $[\text{BH}_2(\text{NH}_3)_2]^+ [\text{BH}_4]^-$
 (3) $[\text{BH}_4]^+ [\text{BH}_2(\text{NH}_3)_2]^-$
 (4) $\text{B}_3\text{N}_3\text{H}_6$
- Which of the following statement is wrong regarding diborane
 (1) It is highly toxic gas
 (2) It is Inflammable
 (3) It is not hydrolysed by water
 (4) It acts as lewis acid
- Al liberate H_2 gas on dissolving it in
 (1) HCl (2) HNO_3
 (3) NaOH (4) 1 & 3 both

CARBON FAMILY

- Elements of group 14
 (1) exhibit oxidation state of +4 only
 (2) exhibit oxidation state of +2 and +4
 (3) form M^{2-} and M^{+4} ions
 (4) form M^{+2} and M^{+4} ions.
- Normally SiO_2 is almost non reactive due to
 (1) Very high Si–O bond enthalpy
 (2) Back bonding between Si & O atom
 (3) sp^3 hybridisation of Si
 (4) None of these
- SiO_2 is soluble in
 (1) HNO_3 (2) H_2SO_4 (3) HF (4) HCl
- The chain length of silicones can be controlled by
 (1) $(\text{CH}_3)_3\text{SiCl}$
 (2) Addition of Cu powder
 (3) Elevation of temperature
 (4) None of these
- Carbon monoxide does not act as
 (1) Reducing agent
 (2) Complexing agent
 (3) Fuel
 (4) A basic oxide

17. In graphite C – C bond length (141.5 pm) found to be shorter than normal C – C bond length (154 pm) this anomaly occurs due to
- (1) There is $\pi\pi$ - $\pi\pi$ bond delocalise within layer
 - (2) In Hexagonal layer structure C-atom are bonded more compactly.
 - (3) Hexagonal layers have weak vander waals forces among them.
 - (4) sp^3 hybridisation of each carbon atom.

NITROGEN FAMILY

18. Which of the following can form $d\pi$ - $d\pi$ bond with transition metal?
- (1) $N(CH_3)_3$
 - (2) $P(C_2H_5)_3$
 - (3) $As(C_6H_5)_3$
 - (4) 2 & 3 both
19. Nitrogen form NX_3 type trihalide. Most stable trihalide is–
- (1) NF_3
 - (2) NCl_3
 - (3) NBr_3
 - (4) NI_3
20. BiH_3 is strongest reducing agent among hydride of 15th group element due to
- (1) Most ionic hydride
 - (2) least bond dissociation enthalpy
 - (3) smallest bond angle
 - (4) Highest boiling point
21. PH_3 is non-inflammable gas, but becomes inflammable owing to the presence of
- (1) P_2H_4
 - (2) P_4O_{10}
 - (3) NaH_2PO_2
 - (4) $NaOH$
22. In the gaseous state of HNO_3 N – $\hat{O}$ – N bond angle is 102° what would be the arrangement of HNO_3
- (1) Planar
 - (2) Tetrahedral
 - (3) Square planar
 - (4) Distorted
23. PH_3 is slightly soluble in water, the solution of PH_3 in water decomposes in presence of light and gives
- (1) $PH_4^+OH^-$
 - (2) H_2 & P_4 (red)
 - (3) P_4O_{10}
 - (4) P_2H_4 + energy
24. Which of the following oxoacid of phosphorus does not disproportionate?
- (1) H_3PO_2
 - (2) H_3PO_4
 - (3) H_3PO_3
 - (4) All
25. The electronegativity of N & Cl are almost equal but N_2 is inert while Cl_2 is reactive because
- (1) Cl – Cl bond has very high bond dissociation energy
 - (2) N_2 molecule has very high bond dissociation energy
 - (3) Cl_2 molecule is bigger in size
 - (4) N has +ve electron gain enthalpy
26. PCl_3 is an oily colorless liquid, it hydrolyses in presence of moisture and produces–
- (1) H_3PO_3 & HCl
 - (2) H_3PO_4 & HCl
 - (3) H_3PO_2 & H_3PO_3
 - (4) H_3PO_3 & H_3PO_4
27. In the Ostwald's process nitric acid is prepared by the catalytic oxidation of
- (1) N_2
 - (2) NH_3
 - (3) N_2O_5
 - (4) NO_2
28. Holme signal contains
- (1) CaC_2 + $CaCN_2$
 - (2) CaC_2 + Ca_3P_2
 - (3) CaC_2 + $CaCO_3$
 - (4) Ca_3P_2 + $CaCN_2$

OXYGEN FAMILY

29. What is the correct order of acidic nature?
- (1) $SO_2 < SeO_2 < TeO_2$
 - (2) $SO_2 > SeO_2 > TeO_2$
 - (3) $SO_2 = SeO_2 = TeO_2$
 - (4) $SO_2 < SeO_2 > TeO_2$
30. Which of the following oxide is different from others?
- (1) SO_2
 - (2) Cl_2O_7
 - (3) Pb_3O_4
 - (4) CO_2
31. For the estimation of O_3 , it is allowed to react with excess of KI solution buffered with a
- (1) Borate buffer
 - (2) NH_4OH + NH_4^+ buffer
 - (3) CH_3COOH + CH_3COONa buffer
 - (4) All
32. Which of the following form of sulphur is paramagnetic?
- (1) S_3
 - (2) S_6
 - (3) S_2
 - (4) All of them
33. What are most favourable conditions for the production of SO_3 from SO_2 ?
- (1) High temp & low pressure
 - (2) low temp & high pressure
 - (3) low temp & low pressure
 - (4) High temp & high pressure

HALOGEN FAMILY

- 34.** Interhalogen compounds are—
(1) Ionic compound
(2) Coordinate compound
(3) Molecular solid
(4) Covalent compound
- 35.** In the oxyacids of chlorine Cl – O bond contains—
(1) $d\pi - d\pi$ bonding (2) $p_\pi - p_\pi$ bonding
(3) $p_\pi - p_\pi$ bonding (4) None
- 36.** Which of the following oxide of chlorine is used for the bleaching of paper pulp & textiles?
(1) Cl_2O (2) Cl_2O_6
(3) Cl_2O_7 (4) ClO_2
- 37.** Three part of conc. HCl & one part of HNO_3 are mixed, aquaregia is formed, it is used to dissolve
(1) Inert gases (2) Noble metals
(3) Alkali metal (4) All

INERT GASES

- 38.** Which of the following gas found in Pitchblende & monazite mineral?
(1) Ar (2) He
(3) Kr (4) All
- 39.** Which of the following statement is wrong regarding Noble gases?
(1) They are readily soluble in water
(2) Interatomic interaction in these element is weak dispersion force
(3) Inert gases have the property of diffusing through rubber and glass etc.
(4) MP & B.P. of these element is quite lower
- 40.** Which of the following gas is used in magnetic resonance imaging (MRI)?
(1) Ni (2) Liq. He
(3) Xe (4) Rn